

Probability – II

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References:

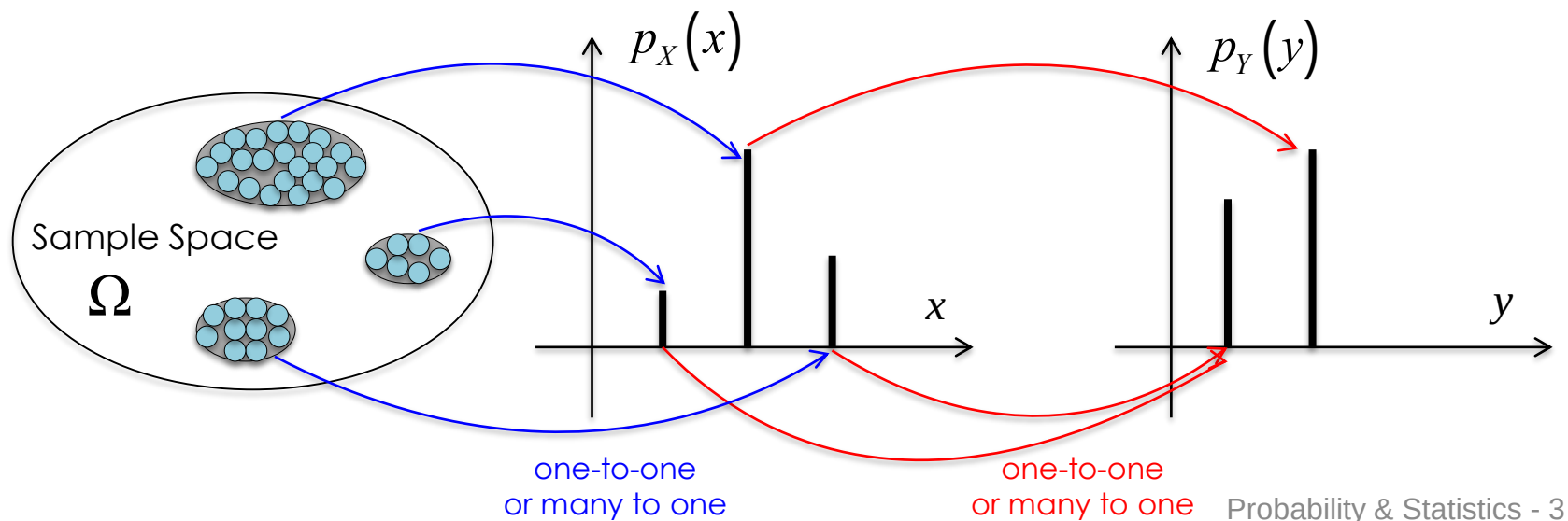
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- Slides are credited from Prof. Berlin Chen, NTNU.

LINEAR FUNCTIONS OF RANDOM VARIABLES

Functions of Random Variables (1/2)

- Given a random variable X , other random variables can be generated by applying various transformations on X
 - Linear $Y = g(x) = aX + b$

↗ Daily temperature in degree Fahrenheit
 ↖ Daily temperature in degree Celsius
 - Nonlinear $Y = g(x) = \log X$



Functions of Random Variables (2/2)

- That is, if Y is an function of X ($Y = g(X)$), then Y is also a random variable
 - If X is discrete with PMF $p_X(x)$, then Y is also discrete and its PMF can be calculated using

$$p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$$

Functions of Random Variables: An Example

Example 2.1. Let $Y = |X|$ and let us apply the preceding formula for the PMF p_Y to the case where

$$p_X(x) = \begin{cases} 1/9 & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0 & \text{otherwise.} \end{cases}$$

The possible values of Y are $y = 0, 1, 2, 3, 4$. To compute $p_Y(y)$ for some given value y from this range, we must add $p_X(x)$ over all values x such that $|x| = y$. In particular, there is only one value of X that corresponds to $y = 0$, namely $x = 0$. Thus,

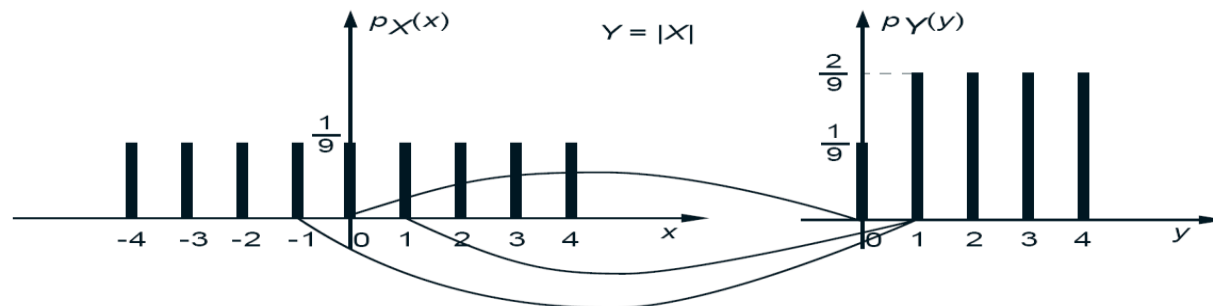
$$p_Y(0) = p_X(0) = \frac{1}{9}.$$

Also, there are two values of X that correspond to each $y = 1, 2, 3, 4$, so for example,

$$p_Y(1) = p_X(-1) + p_X(1) = \frac{2}{9}.$$

Thus, the PMF of Y is

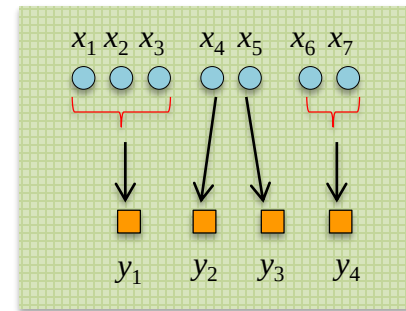
$$p_Y(y) = \begin{cases} 2/9 & \text{if } y = 1, 2, 3, 4, \\ 1/9 & \text{if } y = 0, \\ 0 & \text{otherwise.} \end{cases}$$



Expectations for Functions of Random Variables

- Let X be a random variable with PMF p_X , and let $g(X)$ be a function of X . Then, the expected value of the random variable $g(X)$ is given by

$$\mathbf{E}[g(X)] = \sum_x g(x) p_X(x)$$



- To verify the above rule
 - Let $Y = g(X)$, and therefore $p_Y(y) = \sum_{\{x|g(x)=y\}} p_X(x)$

$$\begin{aligned} \mathbf{E}[g(X)] &= \mathbf{E}[Y] = \sum_y y p_Y(y) \\ &= \sum_y y \sum_{\{x|g(x)=y\}} p_X(x) = \sum_y \sum_{\{x|g(x)=y\}} g(x) p_X(x) \\ &= \sum_x g(x) p_X(x) \end{aligned}$$

?

Example

- **Example 2.3:** For the a random variable X with PMF

$$p_X(x) = \begin{cases} 1/9, & \text{if } x \text{ is an integer in the range } [-4, 4], \\ 0, & \text{otherwise} \end{cases}$$

Discrete Uniform Random Variable

$$\mathbf{E}(X) = \sum_x x p_X(x) = \frac{1}{9} \sum_{x=-4}^4 x = 0$$

$$\text{var}(X) = \mathbf{E}[(X - \mathbf{E}[X])^2] = \sum_x (x - \mathbf{E}[X])^2 p_X(x) = \frac{1}{9} \sum_{x=-4}^4 x^2 = \frac{60}{9}$$

$$\text{Or, let } Z = (X - \mathbf{E}[X])^2 = X^2$$

$$\Rightarrow p_Z(z) = \begin{cases} 2/9, & \text{if } z = 1, 4, 9, 16 \\ 1/9, & \text{if } z = 0 \\ 0, & \text{otherwise} \end{cases}$$

$$\text{var}(X) = \mathbf{E}[Z] = \sum_z z p_Z(z) = \frac{60}{9}$$

Mean and Variance of Linear Functions of Random Variables (1/2)

- Let X be a random variable and let

$$Y = aX + b \quad \text{a linear function of } X$$

where a and b are given scalars

Then,

$$\mathbf{E}[Y] = a\mathbf{E}[X] + b$$

$$\text{var}(Y) = a^2 \text{var}(X)$$

- If $g(X)$ is a linear function of X , then

$$\mathbf{E}[g(X)] = g(\mathbf{E}[X]) \quad \text{How to verify it ?}$$

Mean and Variance of Linear Functions of Random Variables (2/2)

$$\mathbf{E}[Y] = \sum_x (ax + b) p_X(x) = \left[a \sum_x x p_X(x) \right] + \left[b \sum_x p_X(x) \right] = a\mathbf{E}[X] + b$$

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$$\begin{aligned} \text{var}(Y) &= \sum_x (ax + b - \mathbf{E}[aX + b])^2 p_X(x) \\ &= \sum_x (ax + b - a\mathbf{E}[X] - b)^2 p_X(x) \\ &= a^2 \sum_x (x - \mathbf{E}[X])^2 p_X(x) \\ &= a^2 \text{var}(X) \end{aligned}$$

Linear Functions of Random Variables

- If X is a random variable, and a and b are constants, then

$$\mu_{aX+b} = a\mu_X + b$$

$$\sigma_{aX+b}^2 = a^2 \sigma_X^2$$

$$\sigma_{aX+b} = |a| \sigma_X$$

More Linear Functions

- If X and Y are random variables, and a and b are constants, then

$$\mu_{aX+bY} = \mu_{aX} + \mu_{bY} = a\mu_X + b\mu_Y$$

- More generally, if $X_1, \dots, X_n$ are random variables and $c_1, \dots, c_n$ are constants, then the mean of the linear combination $c_1 X_1, \dots, c_n X_n$ is given by

$$\mu_{c_1 X_1 + c_2 X_2 + \dots + c_n X_n} = c_1 \mu_{X_1} + c_2 \mu_{X_2} + \dots + c_n \mu_{X_n}$$

Variance Properties

- If X and Y are independent random variables with variances , then the variance of the sum $X + Y$ is

$$\sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$$

- The variance of the difference $X - Y$ is

$$\sigma_{X-Y}^2 = \sigma_X^2 + \sigma_Y^2$$

More Variance Properties

- If $X_1, \dots, X_n$ are *independent* random variables, then the variance of the sum $X_1 + \dots + X_n$ is given by

$$\sigma_{X_1 + X_2 + \dots + X_n}^2 = \sigma_{X_1}^2 + \sigma_{X_2}^2 + \dots + \sigma_{X_n}^2$$

- If $X_1, \dots, X_n$ are *independent* random variables and $c_1, \dots, c_n$ are constants, then the variance of the *linear* combination $c_1 X_1 + \dots + c_n X_n$ is given by

$$\sigma_{c_1 X_1 + c_2 X_2 + \dots + c_n X_n}^2 = c_1^2 \sigma_{X_1}^2 + c_2^2 \sigma_{X_2}^2 + \dots + c_n^2 \sigma_{X_n}^2$$

Independence and Simple Random Samples

- Definition: If $X_1, X_2, \dots, X_n$ is a **simple random sample**, then $X_1, X_2, \dots, X_n$ may be treated as independent random variables, all from the same population

- Phrased another way, $X_1, X_2, \dots, X_n$ **are independent, and identically distributed** (i.i.d.)

- Sample Mean:

$$\bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

- Sample Variance:

$$s^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2 \Rightarrow s^2 = \frac{1}{\boxed{n-1}} \sum_{i=1}^n (X_i - \bar{X})^2$$

Properties of $\bar{X}$ (1/4)

- If $X_1, X_2, \dots, X_n$ is a simple random sample from a population with mean μ and variance σ^2 , then the sample mean $\bar{X}$ is a **random variable** with

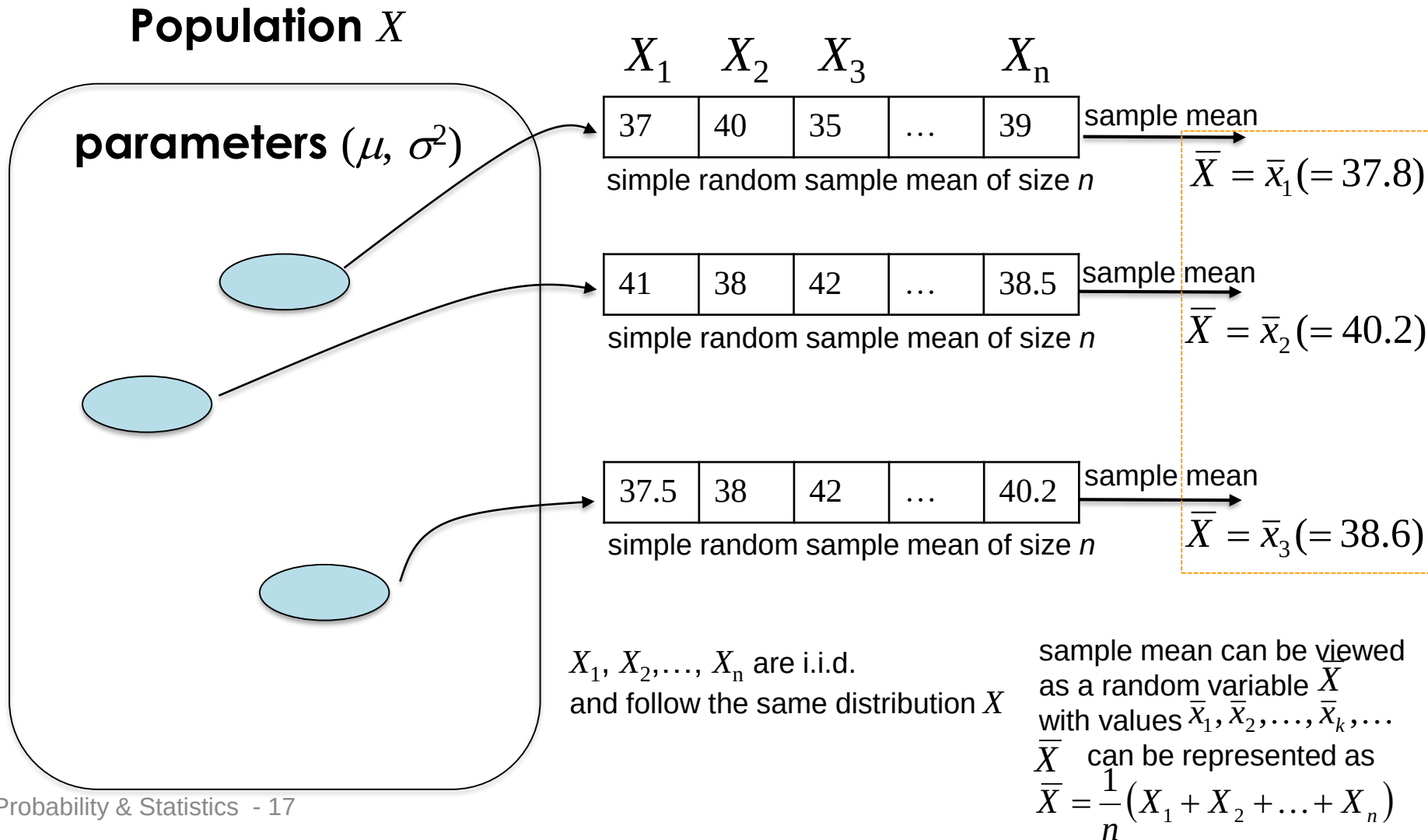
mean of sample mean $\mu_{\bar{X}} = \mu$

variance of sample mean $\sigma_{\bar{X}}^2 = \frac{\sigma^2}{n}$

The standard deviation of $\bar{X}$ is

$$\sigma_{\bar{X}} = \frac{\sigma}{\sqrt{n}}$$

Properties of $\bar{X}$ (2/4)



Properties of $\bar{X}$ (3/4)

$$\mu_{\bar{X}} = \mu_{\frac{1}{n}(X_1 + X_2 + \dots + X_n)}$$

$$= \frac{1}{n} \mu_{X_1} + \frac{1}{n} \mu_{X_2} + \dots + \frac{1}{n} \mu_{X_n}$$

$$= \frac{1}{n} \mu + \frac{1}{n} \mu + \dots + \frac{1}{n} \mu$$

$$= \mu$$

$X_1, X_2, \dots, X_n$ are i.i.d.
and follow the same distribution X with mean μ

$$\sigma_{\bar{X}}^2 = \sigma_{\frac{1}{n}(X_1 + X_2 + \dots + X_n)}^2$$

$$= \frac{1}{n^2} \sigma_{X_1}^2 + \frac{1}{n^2} \sigma_{X_2}^2 + \dots + \frac{1}{n^2} \sigma_{X_n}^2$$

$$= \frac{1}{n^2} \sigma^2 + \frac{1}{n^2} \sigma^2 + \dots + \frac{1}{n^2} \sigma^2$$

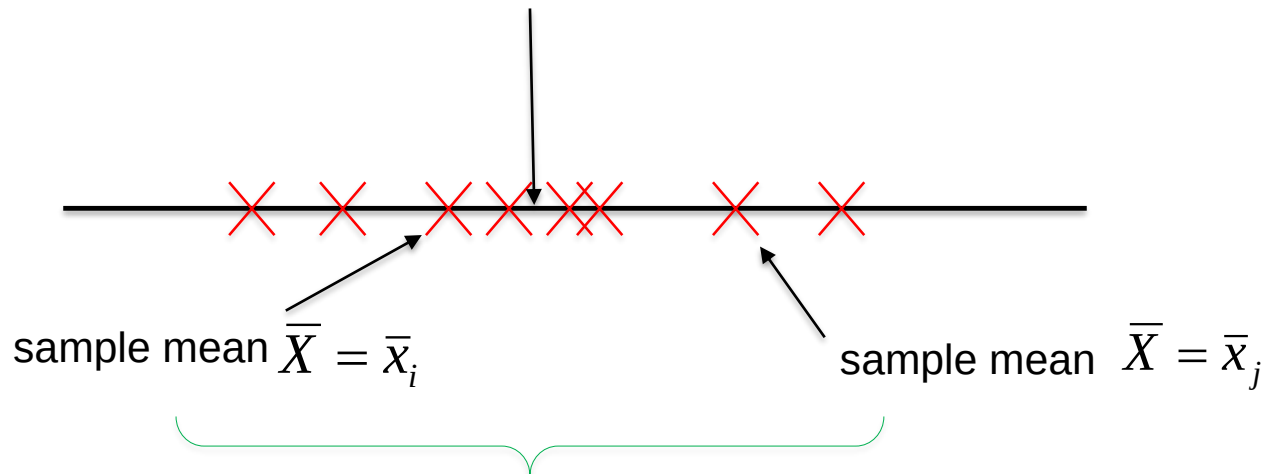
$$= \frac{\sigma^2}{n}$$

$X_1, X_2, \dots, X_n$ are independent

$X_1, X_2, \dots, X_n$ are identically distributed
(follow the same distribution X with variance σ^2)

Properties of $\bar{X}$ (4/4)

mean of sample mean $\mu_{\bar{X}}$ (equal to population mean μ)



The spread of sample mean is determined by the variance of sample mean $\sigma_{\bar{X}}^2$ (equal to $\frac{\sigma^2}{n}$ where σ^2 is the population variance)

INDEPENDENCE – BASICS

Independence of a Random Variable from an Event

- A random variable X is **independent of an event** A if

$$\mathbf{P}(X=x \text{ and } A) = \mathbf{P}(X=x)\mathbf{P}(A), \quad \text{for all } x$$

- Require two events $\{X=x\}$ and A be independent for all x
- If a random variable X is **independent of an event** A and $\mathbf{P}(A) > 0$

$$\begin{aligned} p_{X|A}(x) &= \frac{\mathbf{P}(X=x \text{ and } A)}{\mathbf{P}(A)} \\ &= \frac{\mathbf{P}(X=x)\mathbf{P}(A)}{\mathbf{P}(A)} \\ &= \mathbf{P}(X=x) \\ &= p_X(x), \text{ for all } x \end{aligned}$$

Example

- **Example 2.19.** Consider two independent tosses of a fair coin.
 - Let random variable X be the number of heads
 - Let random variable Y be 0 if the first toss is head, and 1 if the first toss is tail
 - Let A be the event that the number of head is even
 - Possible outcomes (T,T), (T,H), (H,T), (H,H)

$$p_X(x) = \begin{cases} 1/4, & \text{if } x = 0 \\ 1/2, & \text{if } x = 1 \\ 1/4, & \text{if } x = 2 \end{cases}$$

$$p_{X|A}(x) = \begin{cases} 1/2, & \text{if } x = 0 \\ 0, & \text{if } x = 1 \\ 1/2, & \text{if } x = 2 \end{cases}$$

$$p_{X|A}(x) \neq p_X(x) \Rightarrow X \text{ and } A \text{ are not independent!}$$

$$p_Y(y) = \begin{cases} 1/2, & \text{if } y = 0 \\ 1/2, & \text{if } y = 1 \end{cases}$$

$$p_{Y|A}(y) = \frac{\mathbf{P}(Y=y \text{ and } A)}{\mathbf{P}(A)} = \begin{cases} 1/2, & \text{if } y = 0 \\ 1/2, & \text{if } y = 1 \end{cases}$$

$$\mathbf{P}(A) = 1/2$$

$$p_{Y|A}(y) = p_Y(y) \Rightarrow Y \text{ and } A \text{ are independent!}$$

The Prove on Variance Properties

- If X and Y are **independent** random variables, then

$$\text{var}[X+Y] = \text{var}[X] + \text{var}[Y] \quad \text{is equal to} \quad \sigma_{X+Y}^2 = \sigma_X^2 + \sigma_Y^2$$

- As shown by the following calculation

$$\begin{aligned} \text{var}[X+Y] &= \mathbf{E}[\left((X+Y) - \mathbf{E}[X+Y]\right)^2] \\ &= \mathbf{E}[(X+Y)^2 - 2(X+Y)(\mathbf{E}[X] + \mathbf{E}[Y]) + (\mathbf{E}[X] + \mathbf{E}[Y])^2] \\ &= \left[\sum_{x,y} (x+y)^2 p_{X,Y}(x,y) \right] - 2(\mathbf{E}[X] + \mathbf{E}[Y])\mathbf{E}[X] - 2(\mathbf{E}[X] + \mathbf{E}[Y])\mathbf{E}[Y] \\ &\quad + (\mathbf{E}[X])^2 + 2 \cdot \mathbf{E}[X]\mathbf{E}[Y] + (\mathbf{E}[Y])^2 \\ &= \left[\sum_{x,y} x^2 p_{X,Y}(x,y) \right] + \left[\sum_{x,y} y^2 p_{X,Y}(x,y) \right] + 2 \left[\sum_{x,y} xy p_{X,Y}(x,y) \right] \\ &\quad - (\mathbf{E}[X])^2 - (\mathbf{E}[Y])^2 - 2\mathbf{E}[X]\mathbf{E}[Y] \\ &= (\mathbf{E}[X^2] - (\mathbf{E}[X])^2) + (\mathbf{E}[Y^2] - (\mathbf{E}[Y])^2) = \text{var}(X) + \text{var}(Y) \end{aligned}$$

$$\sigma_{X-Y}^2 = ?$$